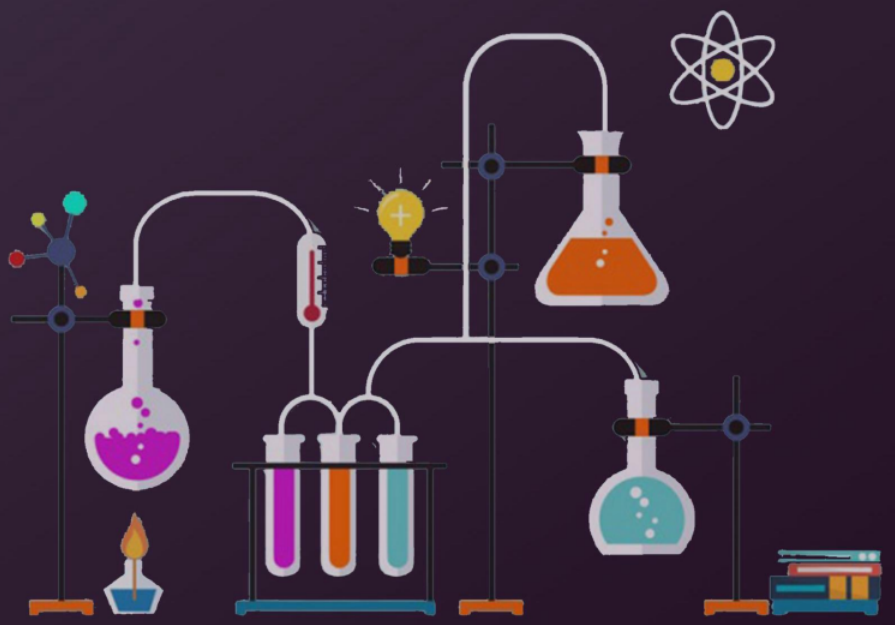
CHEMISTRY

Carbonyl Compound



SHEET

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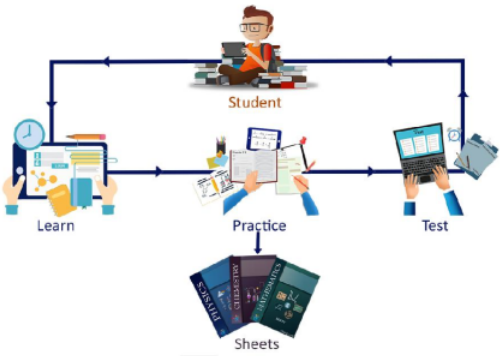
Contents

1.	Module Description	Page - ii
2.	Homework Index	Page - iii
3.	Exercise 1	Page - 1
4.	Exercise 1A	Page - 3
5.	Exercise 2	Page - 6
6.	Exercise 2A	Page - 10
7.	Answer Key	Page - 14

Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topics	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Preparation of Carbonyl Compounds	1-4	1	1-2	
2	Nucleophilic Addition Reaction	5-13	2, 14-15	3-8	14
3	Aldol Reaction and Condensation	14-16	3, 16	9-13	5-6
4	Claisen and Perkin Condensation	17		14	
5	Cannizzaro and Reformatsky Reaction	18-19	4	15-20	1-2, 11, 15
6	Haloform Reaction and Iodoform Test	20	5, 10-11	21-24	3
7	Benzil-Benzilic and Pinacol-Pinacolone Rearrangement				9-10
8	ACC	21-25	6-9, 12-13, 17	25-35	4, 7-8, 12-13

Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Preparation of Carbonyl Compounds

Q1. Which gives acetaldehyde with aqueous KOH

(A) $CH_3-CH(Cl)_2$ (B) $CH_3-C(Cl)_2-CH_3$

(C) $CH_2(Cl)-CH_2(Cl)$ (D) CH_3-CH_2-Cl

Q2. The product (s) obtained via oxymercuration ($HgSO_4 + H_2SO_4$) of 1-butyne would be :-

(A) $CH_3-CH_2-C(=O)-CH_3$

(B) $CH_3 - CH_2 - CH_2 - CHO$

(C) $CH_3 - CH_2 - CHO + HCHO$

(D) $CH_3\ CH_2COOH + HCOOH$

Q3. Gem dihalide on hydrolysis gives :

(A) Vic diol

(B) Gem diol

(C) Carbonyl compound

(D) Carboxylic acid

Q4. Which one of the combinations will give propanaldehyde on dry distillation ?

(A) $(C_6H_5COO)_2Ca$ and $(HCOO)_2Ca$

(B) $(CH_3COO)_2Ca$ and $(CH_3CH_2-COO)_2Ca$

(C) $(CH_3-CH_2-COO)_2Ca$ and $(HCOO)_2Ca$

(D) $(CH_3COO)_2Ca$ and $(CH_3COO)_2Ca$

Nucleophilic Addition Reaction

Q5. The correct order of reactivity of PhMgBr with :

$Ph-C(=O)-Ph$ (I) $CH_3-C(=O)-H$ (II) $CH_3-C(=O)-CH_3$ (III)

(A) I > II > III (B) III > I > II

(C) II > III > I (D) II > I > III

Q6. Rate of the reaction,

$R-C(=O)-Z + Nu^{\ominus} \longrightarrow R-C(=O)-Nu + Z^{\ominus}$

is fastest when Z is :

(A) Cl (B) NH_2

(C) OC_2H_5 (D) $OCOCH_3$

Q7. Rank the following compounds with respect to increasing order of hydration with H_2O, $H^{\oplus}$.

(A) (B)

(C) (D)

(A) D < C < A < B

(B) C < D < A < B

(C) B < A < C < D

(D) A < B < C < D

Q8. The increasing order of the rate of HCN addition to compounds A to D is -

(A) HCHO (B) CH_3COCH_3

(C) $PhCOCH_3$ (D) PhCOPh

(A) D < B < C < A

(B) D < C < B < A

(C) C < D < B < A

(D) A < B < C < D

Q9. Acetaldehyde reacts with semicarbazide, product will be :-

(A) $CH_3-CH=NNHCONH_2$

(B) $CH_3-CH=NCONHNH_2$

(C) $CH_3-CH=NHNH_2$

(D) $CH_3CONHCONH_2$

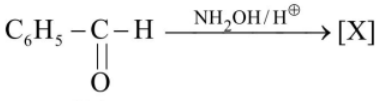
Q10. Arrange these compounds in decreasing order of reactivity for the nucleophilic attack.

(I) Acid chloride (II) Aldehyde
(III) Ketone (IV) Ester

Select the correct answer from the codes given below :

(A) I > II > III > IV (B) IV > III > II > I
(C) III > II > I > IV (D) I > IV > II > III

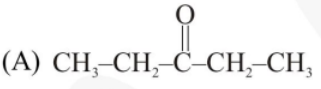
Q11. In the given reaction:

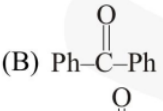


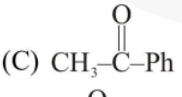
[X] will be:

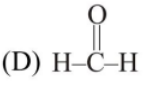
(A) Only syn oxime
(B) Only anti oxime
(C) Mixture of syn and anti oxime
(D) Secondary amide

Q12. Which of the following compound is most reactive when reacts with Grignards reagent (MeMgBr)?

(A) 

(B) 

(C) 

(D) 

Q13. Acetophenone can be obtained by the distillation of :-

(A) $(C_6H_5COO)_2$ Ca
(B) $(CH_3COO)_2Ca$
(C) $(C_6H_5COO)_2Ca$ and $(CH_3COO)_2Ca$
(D) $(C_6H_5COO)_2$ and $(HCOO)_2Ca$

Aldol Reaction and Condensation

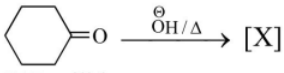
Q14. Propionaldehyde on treatment with dilute NaOH gives :-

(A) $CH_3CH_2COOCH_2CH_2CH_3$
(B) $CH_3CH_2CH(OH)CH(CH_3)CHO$
(C) $CH_3CH_2CHOHCH_2CH_2CHO$
(D) $CH_3CH_2COCH_2CH_2CHO$

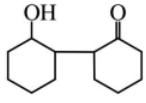
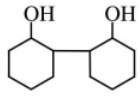
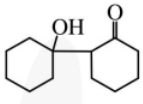
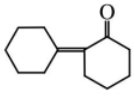
Q15. Which one of the following aldehydes will not form an aldol when treated with dil.NaOH ?

(A) CH_3CHO (B) CH_3CH_2CHO
(C) $(CH_3)_3CCHO$ (D) $C_6H_5CH_2CHO$

Q16. In the reaction :



[X] will be :

(A) 
(B) 
(C) 
(D) 

Claisen and Perkin Condensation

Q17. Product of Perkin reaction is :

(A) α, β-unsaturated aldehyde
(B) β-cyclohexyl α, β-unsaturated aldehyde
(C) α, β-unsaturated acid
(D) All of these

Cannizzaro and Reformatsky Reaction

Q18. In the mechanism of Canizzaro's reaction of HCHO, there is shifting of :

(A) Proton
(B) Hydrogen atom
(C) Hydride ion
(D) Hydronium ion

Q19. Cross cannizzaro reaction is example of :

(A) Redox reaction
(B) Disproportionation
(C) Both (A) and (B)
(D) Only oxidation

Haloform Reaction and Iodoform Test

Q20. A comp. gives a positive test with I_2/NaOH is :-

(A) $CH_3(CH_2)_6CHO$
(B) CH_3–$(CH_2)_6COCH_3$
(C) $CH_3CH_2COCH_2CH_3$
(D) $CH_3(CH_2)_2CHO$

ACC

Q21. Methanol can be distinguish from ethanol by

(A) Sodium (B) Lucas reagent

(C) Tollens reagent (D) I_2 + NaOH

Q22. Number of products (without counting stereoisomers) in the given reaction :

$C_6H_5CHO + CH_3–CHO \xrightarrow{\overset{\Theta}{O}H}$ Product will be :

(A) One (B) Three

(C) Two (D) Four

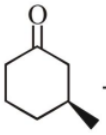
Q23. In the reaction sequence :

$$\text{Cyclohexane} \xrightarrow{hv / Cl_2} (X) \xrightarrow{alc.KOH/\Delta} (Y) \xrightarrow[(ii)\ H_2O/Zn]{(i)\ O_3} (Z)$$

(Z) will be :

(A) Hexanal (B) 2-Hexanone

(C) 3-Hexanone (D) Hexanedial

Q24.  $\xrightarrow[(2)\ H_2O]{(1)\ NaBH_4}$ A + B

Identify relationship between A & B products ?

(A) Diastereoisomers

(B) Enantiomers

(C) Positional isomer

(D) Identical

Q25. Which of the following will give white precipitate with Tollen's reagent.

(A) $CH_3–\overset{\overset{O}{\|}}{C}–H$ (B) $Ph–\overset{\overset{O}{\|}}{C}–H$

(C) $HC\equiv CH$ (D) $H–\overset{\overset{O}{\|}}{C}–OH$

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Hydration of which alkyne give ketone:-

(A) 2-Butyne

(B) Propyne

(C) 1- Butyne

(D) Ethyne

Q2. Two isomeric ketones, 3-pentanone and 2-pentanone can be distinguished by :

(A) I_2 / NaOH

(B) $NaHSO_3$

(C) NaCN / HCl

(D) 2,4-DNP

Q3. Which of the following give aldol reaction with dil. NaOH

(A) CH_3CHO

(B) CD_3CHO

(C) CH_3COCH_3

(D) PhCHO

Q4. $CaC_2 \xrightarrow{H_2O} (P) \xrightarrow[H_2SO_4\,(dil.)]{HgSO_4} (Q)$

$\xrightarrow[\Delta]{dil.\ NaOH} (R) \xrightarrow[ii)\ Zn]{i)\ O_3} (Q) + (S)$

$\xrightarrow[\Delta]{Conc.\ NaOH} (T)$

Correct statement(s) for the above reaction :

(A) (S) to (T) is intramolecular cannizaro

(B) (Q) can show cannizaro reaction

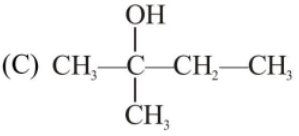
(C) (P) on reaction with Red hot iron tube forms aromatic compound

(D) (R) can show tautomerism

Q5. Which of the following will not form a yellow ppt with I_2 and NaOH.

(A) $Ph—CH_2—CH_2—OH$

(B) $Ph—CH_2—CH(OH)—CH_2—CH_3$

(C) 

(D) $CH_3–CH_2—CH(OH)—CH_3$

Q6. Which of the following reagents can distinguish between propanal and propanone?

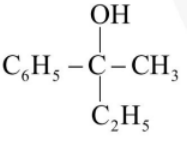
(A) I_2/NaOH

(B) Fehling's solution

(C) Tollens reagent

(D) All of these

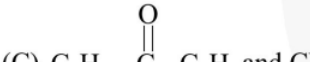
Q7. Consider the structure of given alcohol :



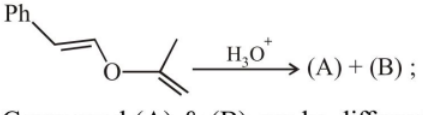
This alcohol can be prepared from:

(A)  and C_2H_5MgBr

(B)  and C_6H_5MgBr

(C)  and CH_3MgBr

(D)  and C_2H_5MgCl

Q8. 

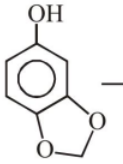
Compound (A) & (B) can be differentiated by ?

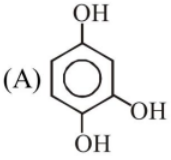
(A) Lucas Reagent

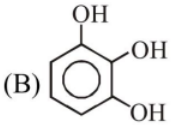
(B) 2,4-DNP Test

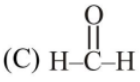
(C) Fehling Solution

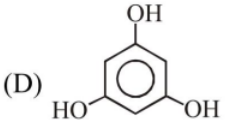
(D) Tollen's Reagent

Q9.  Product.

(A) 

(B) 

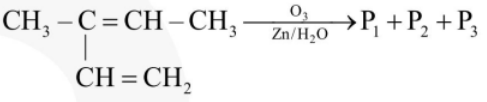
(C) 

(D) 

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) corresponds to the correct answer. For each question, choose the option(s) corresponding to the correct answer.*

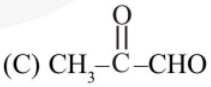
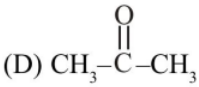
Paragraph for Q. No. 10 & 11



P_1 is a monocarbonyl compound & can not give iodoform test. P_2 is also monocarbonyl compound and will give racemic product with PhMgBr. P_2 can give aldol reaction with dil. NaOH.

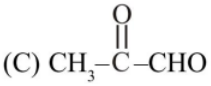
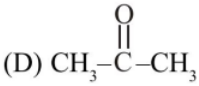
Q10. Identify P_1 from the following.

(A) $CH_3 – CHO$ (B) 

(C)  (D) 

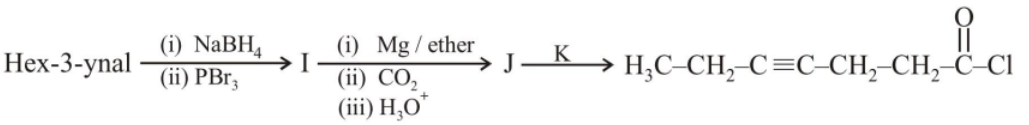
Q11. Identify P_2 from the following.

(A) $CH_3 – CHO$ (B) 

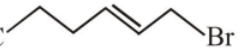
(C)  (D) 

Paragraph for Q.No. 12 to 13

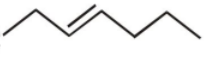
In the following sequence, product I, J and L are formed. K represents a reagent.



Q12. The structure of the product I can't be :

(A) 

(B) 

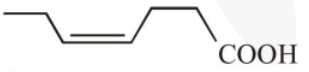
(C) 

(D) $CH_3–CH_2–C\equiv C–CH_2–CH_2–Br$

Q13. The structure of compounds J and K, respectively are :

(A) J is $CH_3–CH_2–C\equiv C–CH_2–CH_2–C(=O)–OH$

(B) K is $SOCl_2$

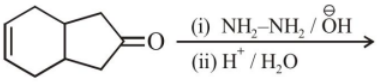
(C) J is 

(D) K is CH_3SO_2Cl

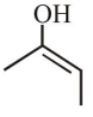
NUMERICAL TYPE

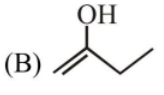
*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

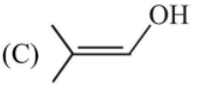
Q14. Degree of unsaturation of major product in following reaction will be ?

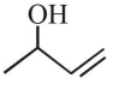


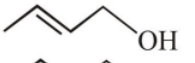
Q15. How many compounds are enol tautomer of 2-butanone.

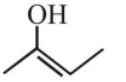
(A) 

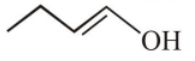
(B) 

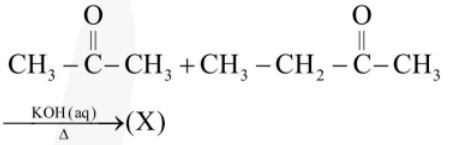
(C) 

(D) 

(E) 

(F) 

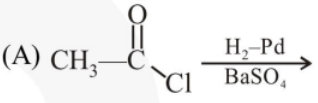
(G) 

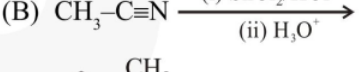
Q16. 

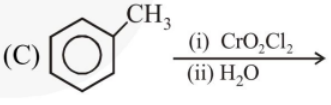
X = Number of Aldol condensation product (including stereoisomer)

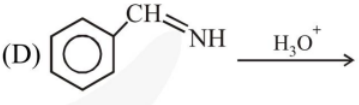
Find out the value of (X).

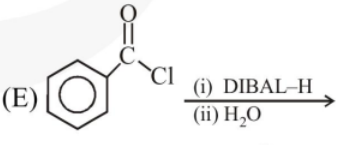
Q17. Number of reactions in which aldehyde is obtained as one of the product.

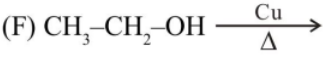
(A) 

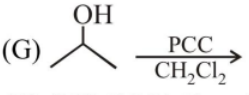
(B) 

(C) 

(D) 

(E) 

(F) 

(G) 

(H) $(CH_3COO)_2Ca + (HCOO)_2Ca \xrightarrow[\text{distillation}]{\text{Dry}}$

(I) $CH_3–MgCl + CH_3–C\equiv N \xrightarrow[H_3O^{\oplus}]{}$

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. The alkene C_6H_{10} producing OHC – $(CH_2)_4$ – CHO on ozonolysis is :-

(A) 1-Hexene

(B) 3-Hexene

(C) Cyclohexene

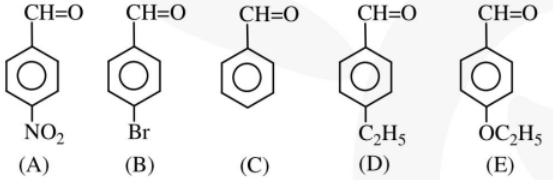
(D) 1-methylcyclohex-1-ene

Q2. Which of the following does not give carbonyl with one mole of grignard reagents followed by hydrolysis:

(A) CO_2 (B) RCOCl

(C) RCN (D) RCOOR

Q3. What is correct order of rate of following carbonyl compound toward hydration :



(A) (B) (C) (D) (E)

(A) A > B > C > D > E

(B) E > D > C > B > A

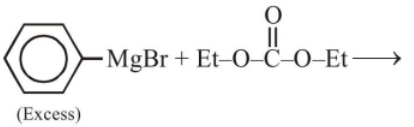
(C) A > C > B > D > E

(D) A > B > E > C > D

Q4. Which of the following is most reactive towards "Nucleophile".

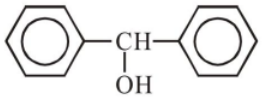
(A) $CH_3-\overset{\overset{O}{\|}}{C}-H$ (B) $F-CH_2-\overset{\overset{O}{\|}}{C}-H$

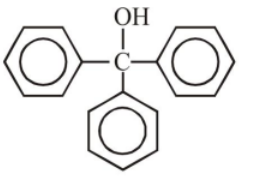
(C) $Cl-CH_2-\overset{\overset{O}{\|}}{C}-H$ (D) All are equal

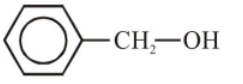
Q5. 

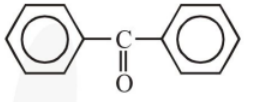
(Excess)

Product, product is ?

(A) 

(B) 

(C) 

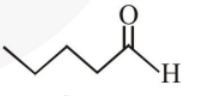
(D) 

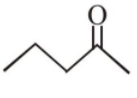
Q6. CH_3 Mg Br (excess) reacts with ethyl ester, then product will :

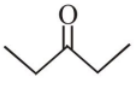
(A) $CH_3-\overset{\overset{C_2H_5}{|}}{C}=O$ (B) $CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{OH}{|}}{C}}-C_2H_5$

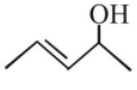
(C) $CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-OH$ (D) $CH_3-CH_2-\overset{\overset{CH_3}{|}}{\underset{\underset{C_2H_5}{|}}{C}}-OH$

Q7. An organic compound (A), $C_5H_{10}O$, reacts with hydrazine to form a hydrazone derivative (B). The hydrazone (B) on being heated with KOH at about 180°C, gives n-pentane. The compound (A) does not respond positively to Tollen's reagent and to the iodoform test. The compound (A) is-

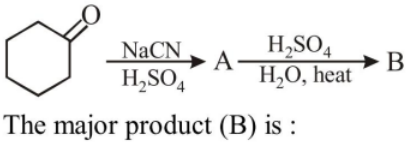
(A) 

(B) 

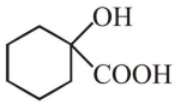
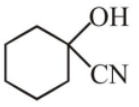
(C) 

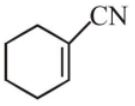
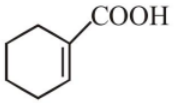
(D) 

Q8. Consider the following sequence of reactions.

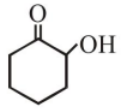
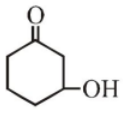


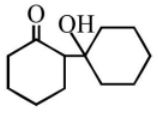
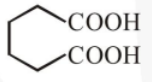
The major product (B) is :

(A)  (B) 

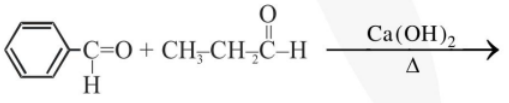
(C)  (D) 

Q9. When cyclohexanone is treated with NaOH solution, we get :-

(A)  (B) 

(C)  (D) 

Q10. Find the major product of given reaction :



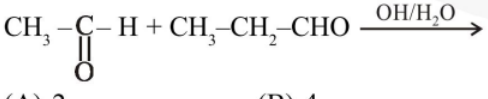
(A) $Ph-CH=CH-CH_2-\overset{O}{\overset{\|}{C}}-H$

(B) $CH_3CH_2-CH=\underset{CH_3}{\overset{|}{C}}-\overset{O}{\overset{\|}{C}}-H$

(C) $CH_3CH_2CH=CH-CH_2-\overset{O}{\overset{\|}{C}}-H$

(D) $Ph-CH=\underset{CH_3}{\overset{|}{C}}-\overset{O}{\overset{\|}{C}}-H$

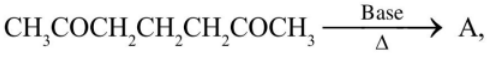
Q11. How many aldol products are possible excluding stereoisomers



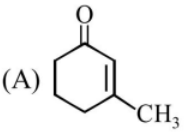
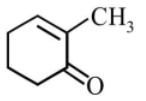
(A) 2 (B) 4

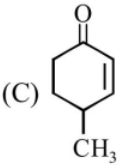
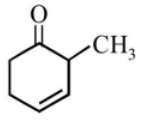
(C) 6 (D) 8

Q12. In the reaction

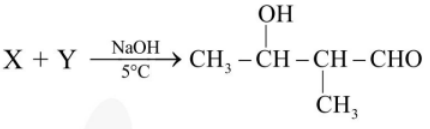


A will be:-

(A)  (B) 

(C)  (D) 

Q13. In the given reaction



(X) and (Y) will respectively be:

(A) CH_3-CH_2-CHO and CH_3-CH_2-CHO

(B) CH_3-CHO and CH_3-CH_2-CHO

(C) CH_3-CHO and CH_3-CHO

(D) CH_3-CHO and $CH_3-\underset{CH_3}{\overset{CH_3}{\overset{|}{\underset{|}{C}}}}-CHO$

Q14. Which of the following does not give Claisen condensation reaction :-

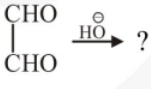
(A) $C_6H_5COOC_2H_5$

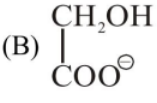
(B) $C_6H_5CH_2COOC_2H_5$

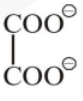
(C) $CH_3COOC_2H_5$

(D) $CH_3CH_2COOC_2H_5$

Q15. The product of the following reaction is



(A)  (B) 

(C)  (D) $CH_3OH + HCOO^{\ominus}$

Q16. A mixture of benzaldehyde and formaldehyde on heating with aqueous NaOH solution gives :-

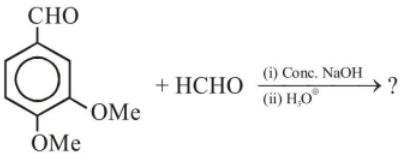
(A) $PhCH_2OH$ and HCOONa

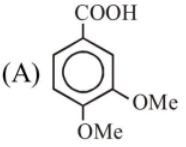
(B) PhCOONa and CH_3OH

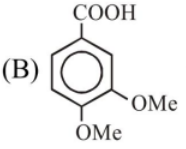
(C) PhCOONa and HCOONa

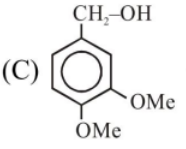
(D) $PhCH_2OH$ and CH_3OH

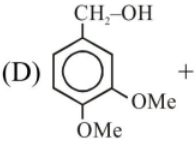
Q17. The following reaction gives :



(A) 

(B) 

(C) 

(D) 

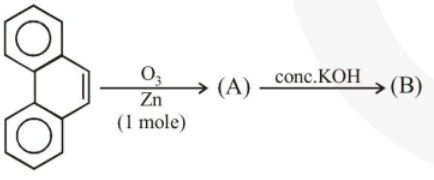
Q18. Phenylglyoxal, C_6H_5COCHO, on heating with concentrated NaOH gives

(A) C_6H_5COONa and CH_3OH

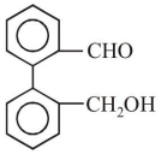
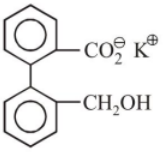
(B) $C_6H_5CH_2OH$ and $HCOONa$

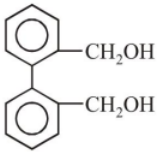
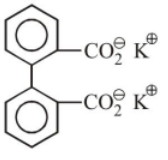
(C) $C_6H_5CHOHCOONa$

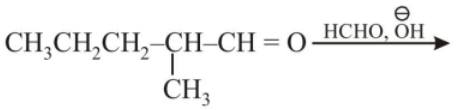
(D) C_6H_5COONa and $HCOONa$

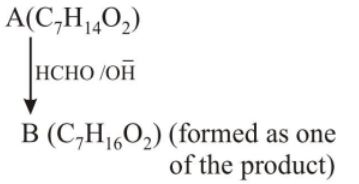
Q19. 

End product (B) of above reaction is :

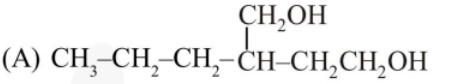
(A)  (B) 

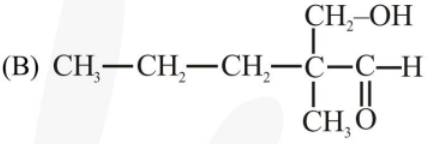
(C)  (D) 

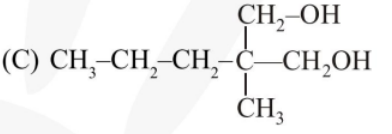
Q20. 



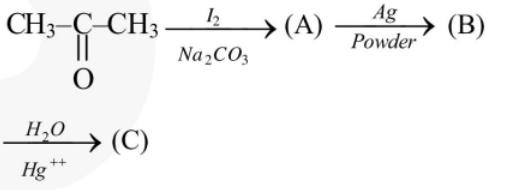
'B' is :

(A) 

(B) 

(C) 

(D) None

Q21. 

Product A, B & C are -

(A) Iodoform, Acetylene & Acetaldehyde

(B) Triiodomethane, Ethyne & Acetone

(C) Iodoform, Ethene & Ethylene glycol

(D) Ethene, iodoform & Ethylhydrogen sulphate

Q22. Which of the following will form a yellow precipitate of iodoform on heating with I_2 and dilute NaOH?

(A) $PhCH_2CH_2OH$

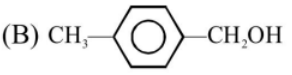
(B) $PhCH_2CHOHCH_2CH_3$

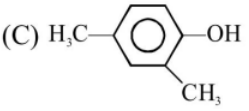
(C) $CH_3CH_2CH_2OH$

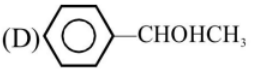
(D) $CH_3CHOHCH_3$

Q23. An organic compound having the molecular formula $C_8H_{10}O$ on being heated with I_2 and dilute NaOH gives a yellow percipitate. The compound is expected to be -

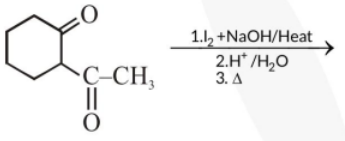
(A) $C_6H_5CH_2CH_2OH$

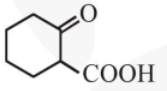
(B) 

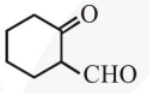
(C) 

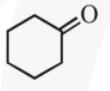
(D) 

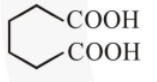
Q24. End products of the following sequence of reactions are :-



(A) yellow ppt. of CHI_3, 

(B) yellow ppt. of CHI_3, 

(C) yellow ppt. of CHI_3, 

(D) yellow ppt. of CHI_3, 

Q25. Which one/ones of the following reactions will yield 2-propanol ?

(i) $CH_3CH = CH_2 + H_2O \xrightarrow{H^+}$

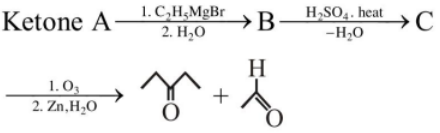
(ii) $CH_3CHO \xrightarrow[(ii)\,H_2O]{(i)\,CH_3MgI}$

(iii) $HCHO \xrightarrow[(ii)H_2O]{(i)\,C_2H_5MgI}$

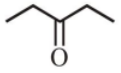
Choose the right answer :

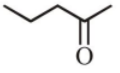
(A) (i) and (ii) (B) (ii) and (iii)
(C) (iii) and (i) (D) None of these

Q26. Consider the following sequence of reactions.



The ketone (A) is -

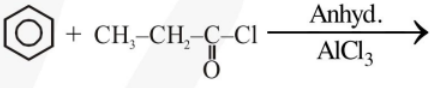
(A)  (B) 

(C)  (D) 

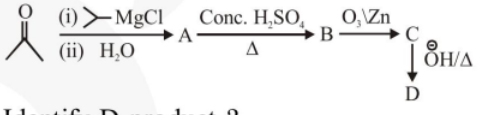
Q27. In which reaction ethyl phenyl ketone is obtained

(A) $C_6H_5–C\equiv N + Et\text{-}MgBr \rightarrow A \xrightarrow[H^{\oplus}]{H_2O}$

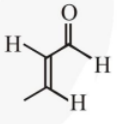
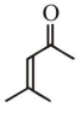
(B) $CH_3–CH_2–\underset{\underset{O}{\|}}{C}–Cl \xrightarrow[H_2O]{Ph_2Cd}$

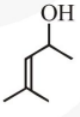
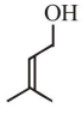
(C) 

(D) All of them

Q28. 

Identify D product ?

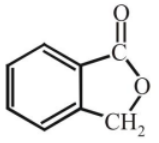
(A)  (B) 

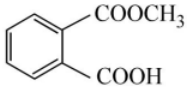
(C)  (D) 

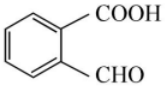
Q29. 2-pentanone can be distinguish from 3-pentanone by the reagent ?

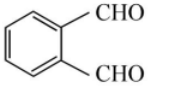
(A) D.N.P.
(B) Tollen's reagent
(C) I_2 and dil NaOH
(D) Fehling's reagent

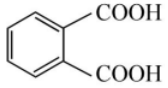
Q30. Which of the following reactants on reaction with conc. NaOH followed by acidification gives the following lactone as the only product



(A) 

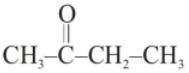
(B) 

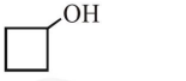
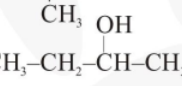
(C) 

(D) 

Q31. Which of the following can be used to differentiate pentan-2-one & pentan-3-one
(A) Iodoform test (B) Fehling reagent
(C) Both 1 & 2 (D) Tollen's reagent

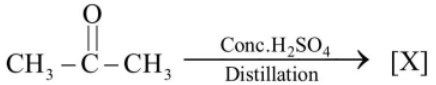
Q32. Compound (X) C_4H_8O, which reacts with 2, 4–DNP derivative and gives negative haloform test is :-

(A)  (B) 

(C)  (D) 

Q33. An organic comp. C_3H_6O does not gives precipitate with bready reagent and does not reacts with metallic sodium, then comp will be :
(A) CH_3COCH_3
(B) $CH_2 = CH - O - CH_3$
(C) $CH_3CH_2 - CHO$
(D) $CH_2 = CH - CH_2 - OH$

Q34. In the given reaction :



[X] will be :
(A) Methyl oxide
(B) Phorone
(C) 1, 3, 5-Trimethylbenzene
(D) 2-Butyne

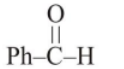
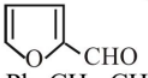
Q35. Acetaldehyde cannot give :
(A) Iodoform test (B) Lucas test
(C) Benedict test (D) Tollens test

Exercise - 2A *Brain Booster*

MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

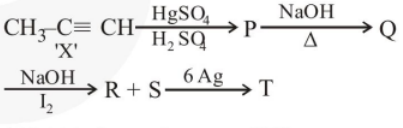
Q1. Which of the following compounds will undergo Canizaro reaction, when treated with 50% aqueous alkali :-

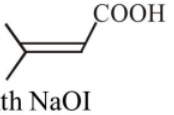
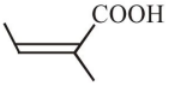
(A)  (B) 

(C) Me_3CCHO (D) $Ph–CH_2–CHO$

Q2. Mixture of Ph–CHO & HCHO is treated with NaOH then Cannizzaro reaction involves:
(A) Oxidation of HCHO
(B) Reduction of HCHO
(C) Oxidation of Ph–CHO
(D) Reduction of Ph–CHO

Q3. Correct option regarding following reaction sequence.



(A) T is homologue of 'X'
(B) R is sodium salt of 
(C) P can produce S with NaOI
(D) R is sodium salt of 

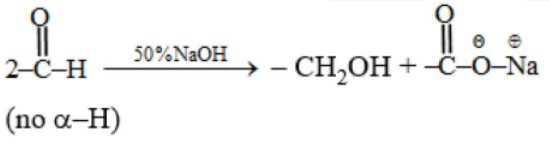
Q4. Which is true about acetophenone :-
(A) Reacts to form 2,4-dinitorphenyl hydrazone
(B) Reacts with Tollen's reagent to form silver mirror
(C) Reacts with I_2/NaOH to form iodoform
(D) On oxidation with alkaline $KMnO_4$ followed by hydrolysis gives benzoic acid

PARAGRAPH TYPE

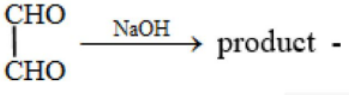
*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) corresponds to the correct answer. For each question, choose the option(s) corresponding to the correct answer.*

Paragraph for Q.No. 5 to 6

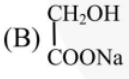
Aldehyde without a–H atoms on reaction with concentrated alkali solution undergo an oxidation-reduction (disproportionation) reaction. One half of the aldehyde is reduced to a 1° alcohol and the other is oxidised to a carboxylic acid. This is known as Cannizzaro reaction.



Q5. The product formed in the following reaction will be -



(A) 

(B) 

(C) 

(D) All of these

Q6. Mixture of C_6H_5CHO and HCHO is treated with NaOH then cannizzaro's reaction involves-

(A) Oxidation of HCHO

(B) Reduction of HCHO and Oxidation of C_6H_5CHO

(C) Reduction of C_6H_5OH

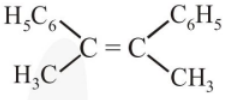
(D) Both A & C

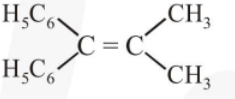
Paragraph for Q.No. 7 to 8

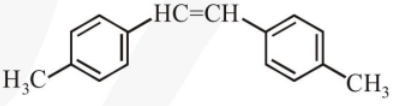
An alkene X ($C_{16}H_{16}$) on ozonolysis gives only one product Y (C_8H_8O) on reaction with NaOH / I_2 yields sodium benzoate. Further Y reacts with KOH / NH_2–NH_2 yielding C_8H_{10}.

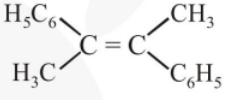
The alkene X has two isomeric strcutures and one isomer (Z) on catalytic hydrogenation (H_2 / Pd–C) yields a racemic mixture.

Q7. The isomer Z can't be :

(A) 

(B) 

(C) 

(D) 

Q8. An isomer of compound Y is reacted further with one mole of methyl magnesium bromide and then heated in acid to give an alkene. The alkene on ozonolysis gives two compound P and Q can be differentiated by :

(A) Fehling solution

(B) Iodoform test

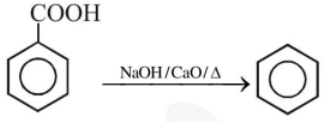
(C) Benedict's solution

(D) Lucas reagent

MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LISTII**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q9. Match list-I with list-II

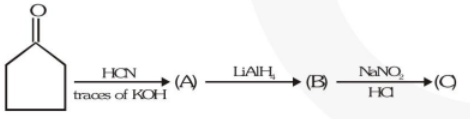
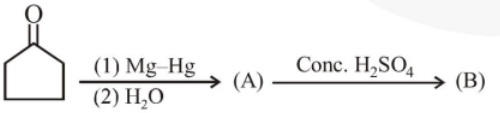
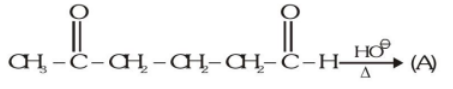
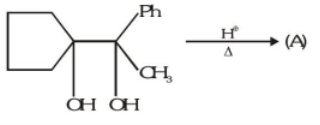
List-I	List-II
(P) $C_6H_5CHO + HCHO \xrightarrow{\overset{\ominus}{O}H} C_6H_5CH_2OH + HCO\overset{\ominus}{O}$	(1) Cannizzaro reaction
(Q) $CH_3-\overset{O}{\overset{\|}{C}}-H \xrightarrow[\Delta]{OH^-} CH_3-CH=CH-CHO$	(2) Aldol condensation
(R) $Ph-\overset{O}{\overset{\|}{C}}-\overset{O}{\overset{\|}{C}}-Ph \xrightarrow[H_2O]{OH^-} Ph-\underset{Ph}{\overset{OH}{\underset{\|}{C}}}-\overset{O}{\overset{\|}{C}}-OH$	(3) Soda-lime Decarboxylation
(S) 	(4) Benzil benzilic acid rearragement

	P	Q	R	S
(A)	1	3	2	4
(B)	1	2	4	3
(C)	4	2	1	3
(D)	3	4	1	2

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

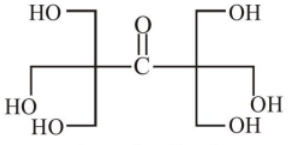
Q10.

Column - I	Column - II
(A) 	(P) Formation of six member ring takes place
(B) 	(Q) Final product is Ketone
(C) 	(R) Final product formed will give positive Idoform test
(D) 	(S) Final pr oduct formed will react with 2,4-DNP.(2,4-Di-nitrophenyl hydrazine)

NUMERICAL TYPE

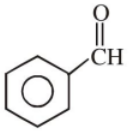
*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

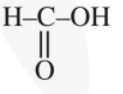
Q11. $CH_3-\overset{O}{\overset{\|}{C}}-CH_3 + xHCHO \xrightarrow{KOH}$

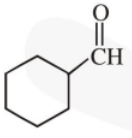


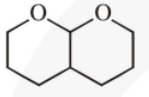
x = moles of HCHO consumed value of (x) will be :

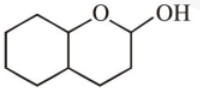
Q12. Number of compounds which can be oxidised by Tollen's regent.

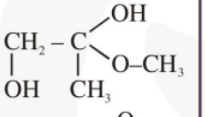
(i) 

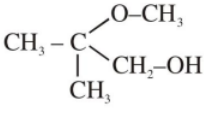
(ii) 

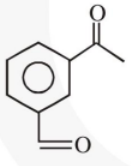
(iii) 

(iv) 

(v) 

(vi) 

(vii) 

(viii) 

Q13. $R-\overset{O}{\overset{\|}{C}}-R \xrightarrow{?} R-CH_2-R$

Identify numbers of reagent that can be used for above conversion.

(a) Zn–Hg / HCl (b) $LiAlH_4$

(c) $CHCl_3$ + NaOH

(d) $\begin{matrix} CH_2-SH \\ | \\ CH_2-SH \end{matrix} \Big/ H_2-Ni$ (e) SeO_2

(f) N_2H_4 / ^-OH

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q14. **Assertion** : Benzaldehyde with HCN gives two isomeric compounds.

Reason : Both nitrile and isonitrile compounds are possible when HCN reacts with carbonyl group.

Q15. **Assertion :** $Cl_3C-\overset{O}{\overset{\|}{C}}-H \xrightarrow{NaOH}$

$Cl_3C—CH_2OH + Cl_3C—COONa.$

Reason : There are no α–H in this compound, so it can't give aldol.

Answer Key

Ex-1

1	2	3	4	5	6	7	8	9	10
A	A	C	C	C	A	C	B	A	A
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
C	D	C	B	C	D	C	C	A	B
21	**22**	**23**	**24**	**25**					
D	C	D	A	C					

Ex-1A

1. A,B,C **2.** A,B **3.** A,B,C **4.** A,C,D **5.** A,B,C

6. D **7.** A,B,C **8.** C,D **9.** A,C **10.** B

11. A **12.** A,B,C **13.** A,B **14.** 2 **15.** 3

16. 6 **17.** 7

Ex-2

1	2	3	4	5	6	7	8	9	10
C	A	A	B	B	C	C	D	C	D
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
B	A	B	A	B	A	D	C	B	C
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
A	D	D	C	A	B	D	B	C	C
31	**32**	**33**	**34**	**35**					
C	B	B	C	B					

Ex-2A

1. A,B,C **2.** A,D **3.** A,B,C **4.** A,C,D **5.** B

6. D **7.** A,B,C **8.** A,C A,B,C **9.** B

10. (A) P,Q,S ; (B) P,Q,S ; (C) P,Q,S ; (D) P,Q,S **11.** 6 **12.** 6

13. 3 **14.** C **15.** D